

One Blind Rotation, Four Output Bits: a Toy Study of Multi-Value Bootstrapping for 4-bit S-boxes in TFHE (teaching example — not a research result)

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Abstract. Programmable bootstrapping (PBS) evaluates any look-up table on a TFHE ciphertext, but one PBS returns one output; a 4-bit S-box whose output bits are consumed separately therefore costs four blind rotations [6]. Multi-value bootstrapping [2] (CT-RSA’19) shares one blind rotation among several tables of the same input.

We re-derive the factorisation $T_f = v_0 \cdot d_f$ for the standard one-padding-bit encoding, show that d_f has at most p non-zero integer coefficients with $\|d_f\|^2 \leq p + 2$, and give the resulting output variance $\|d_f\|^2 \sigma_{\text{BR}}^2 + \sigma_{\text{KS}}^2$. In a pure-Python toy TFHE (insecure parameters, single thread, CPU time), the PRESENT S-box costs 75.4 ms instead of 291.2 ms, a $3.86 \times [3.79, 3.95]$ speed-up, which saturates for many tables ($43.67 \times$ for 64 tables). The price is noise: at a deliberately noisy parameter set the S-box failure rate rises from 0/128 to 32/128. All code, logs and falsification scripts are part of the case study.

1 Introduction

TFHE [3,5] refreshes a ciphertext with a *blind rotation* of a test polynomial, a technique going back to FHEW [7]. Choosing the test polynomial evaluates an arbitrary function during the refresh (programmable bootstrapping, PBS) [6]. The blind rotation dominates the cost, and it does not depend on the function: evaluating k functions of the *same* input with k independent PBS repeats it k times.

A 4-bit S-box is the textbook instance. If the next layer is bit-sliced, the four output bits are needed as four ciphertexts, i.e. four PBS. (If a single ciphertext of the nibble suffices, one PBS with $f = S$ does the job and there is nothing to optimise.) Carpov, Izabachène and Mollimard [2] observed that the test polynomials of many functions share a common factor v_0 , so one blind rotation of v_0 can serve all of them (multi-value PBS, MV-PBS). Tree-based and multi-value variants for larger tables were studied in [8]; CMux-tree evaluation from GGSW inputs goes back to [4].

1.1 Contributions (of this toy study)

- A self-contained derivation of the factorisation for the one-padding-bit encoding (Lemma 1), the exact bound $\|d_f\|^2 \leq p + 2$ for Boolean tables (Lemma 2), and the output-variance formula (Theorem 1).

- A falsification-first evaluation: $3.86\times$ for the PRESENT S-box, sub-linear scaling in the number of tables, and a parameter set where the noise price becomes visible.

We make no novelty claim: everything here is known since [2].

2 Preliminaries

Let $q = 2^{32}$, $R = \mathbb{Z}[X]/(X^N + 1)$ with N a power of two, and $\mathcal{R}_q = R/qR$. Messages $m \in \mathbb{Z}_p$ are encoded with one padding bit as Δm , $\Delta = q/(2p)$; decoding fails when the error reaches $1/(4p)$ of the torus, i.e. 2^{-6} for $p = 16$. A PBS for $f : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ blind-rotates the test polynomial $T_f = \sum_{j < N} \Delta f(\lfloor jp/N \rfloor) X^j$ by the mod-switched phase (with a half-box offset), extracts coefficient 0 as an LWE ciphertext and key-switches it back to the input key [5,6]. We write σ_{BR}^2 and σ_{KS}^2 for the variances added by blind rotation and key switching.

Toy parameters. $n = 32$, $N = 512$, $k = 1$, gadget base 2^7 with 3 levels, key-switch base 2^4 with 5 levels, LWE $\sigma = 2^{-20}$; set T1 uses GLWE $\sigma = 2^{-28}$, the noisy set T2 uses $\sigma = 2^{-22}$. Both are insecure.

3 Multi-Value PBS for Bit-Sliced S-boxes

Lemma 1 (factorisation, cf. [2]). *Let $p \mid N$ and Δ be even. For $f : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ let $d_f \in \mathbb{Z}[X]$ have $d_f[0] = f(0) + f(p-1)$, $d_f[iN/p] = f(i) - f(i-1)$ for $1 \leq i < p$ and zeros elsewhere, and let $v_0 = \frac{\Delta}{2} \sum_{j < N} X^j$. Then $T_f = v_0 \cdot d_f$ in $\mathcal{R}_q$.*

Proof. In R , $(1-X) \sum_{j < N} X^j = 1 - X^N = 2$, so $v_0(1-X) = \Delta$ and $(1-X)$ is a unit after inverting 2. Coefficient $j \geq 1$ of $T_f(1-X)$ is $t_j - t_{j-1}$, non-zero only at block boundaries, and coefficient 0 is $t_0 + t_{N-1}$ by negacyclicity; hence $T_f(1-X) = \Delta d_f = v_0(1-X)d_f$; cancelling the unit $(1-X)$ gives $T_f = v_0 d_f$. All coefficients of v_0 are the integer $\Delta/2$, so the identity holds modulo q .

Since blind rotation is R -linear in the test polynomial, rotating v_0 once and multiplying the accumulator by each d_{f_i} afterwards yields the rotations of all T_{f_i} : MV-PBS costs one blind rotation plus, per table, one product with a polynomial of at most p small non-zero integer coefficients, one sample extraction and one key switch. We checked the identity exhaustively for all Boolean tables with $p \leq 16$ and for random tables, and MV-PBS decrypted all 256/256 outputs of our correctness attack (key seeds, all inputs, extreme tables) correctly at T1.

Lemma 2 (norm of d_f). *For Boolean $f : \mathbb{Z}_p \rightarrow \{0,1\}$, $\|d_f\|^2 = (f(0) + f(p-1))^2 + \#\{i : f(i) \neq f(i-1)\} \leq p+2$, and the bound is attained. For the PRESENT S-box [1] the four output bits give $\|d_f\|^2 = 10, 10, 8, 8$.*

Proof. If $f(0) = f(p-1) = 1$ the number of changes along $0, \dots, p-1$ is even, hence at most $p-2$; otherwise $d_f[0]^2 \leq 1$ and there are at most $p-1$ changes.

Table 1. PRESENT S-box, four output-bit ciphertexts, toy set T1 (CPU time per S-box, median of 5 interleaved rounds over all 16 inputs), and S-box failure rate over 128 random evaluations at the noisy set T2.

	time (T1)	speed-up (T1)	S-box failures (T2)
per-bit PBS (4 blind rotations)	291.2 ms	1	0/128
MV-PBS (1 blind rotation)	75.4 ms	$3.86 \times [3.79, 3.95]$	32/128

Theorem 1 (output noise). *Under the usual heuristic that the accumulator noise coefficients are independent and centred, the MV-PBS output for f has variance $\|d_f\|^2 \sigma_{\text{BR}}^2 + \sigma_{\text{KS}}^2$, whereas a PBS output has $\sigma_{\text{BR}}^2 + \sigma_{\text{KS}}^2$.*

Proof (sketch). Multiplication by the plaintext polynomial d_f maps the noise polynomial e to $d_f e$, whose constant coefficient is a fixed signed combination of $\|d_f\|^2$ -weighted independent terms; sample extraction is exact and key switching adds σ_{KS}^2 .

So MV-PBS gives the same correctness as per-bit PBS *only if* $\|d_f\|^2 \sigma_{\text{BR}}^2 + \sigma_{\text{KS}}^2$ still fits the decoding margin. For the PRESENT bits the blind-rotation noise grows by at most 1.66 bits. At T1 the key-switch noise ($2^{-12.13}$) dominates the blind-rotation noise ($2^{-14.47}$) and the growth is invisible; at T2 it is not (Section 4).

4 Toy Evaluation

Setup. All three strategies run in the same toy Python TFHE library with the same keys, encoding and parameters; the baseline is the library’s own PBS, called once per output bit. We time only the homomorphic evaluation (process CPU time, one thread, numpy), with 5 interleaved rounds after one warm-up round and the arm order rotating between rounds; we report medians and paired-bootstrap 95% CI of the ratio of medians. Every timed execution also checks every output bit against the cleartext S-box. The machine is a shared workstation, which is why we use CPU time rather than wall time (wall time is logged too). These are toy numbers: they measure an algorithmic ratio, not the speed of any real library.

Speed. Table 1 gives $3.86 \times$ for the full S-box. Figure 1(a) scans the number of tables k : at $k = 1$ MV-PBS is as fast as PBS ($0.96 \times$), and at $k = 4$ the speed-up is $3.96 \times$, close to k . It is *not* linear in general: with 64 tables of the same nibble (the four S-box bits plus random Boolean tables) we measure $43.67 \times [42.53, 44.63]$, because the per-table work (product with d_f , sample extraction, key switch) does not amortise.

Noise. At T1, 0 of 384 evaluated output bits were wrong (both methods). At T2 the per-bit PBS output has measured standard deviation $2^{-8.54}$ (model $2^{-8.50}$) and never fails (0/128, Wilson upper end 0.029), whereas the MV-PBS output for bit 0 has $2^{-6.74}$ (model $2^{-6.84}$) against a margin of 2^{-6} . The S-box failure rate grows with the number of bits used, from 14/128 at $k = 1$ to 32/128 at $k = 4$,

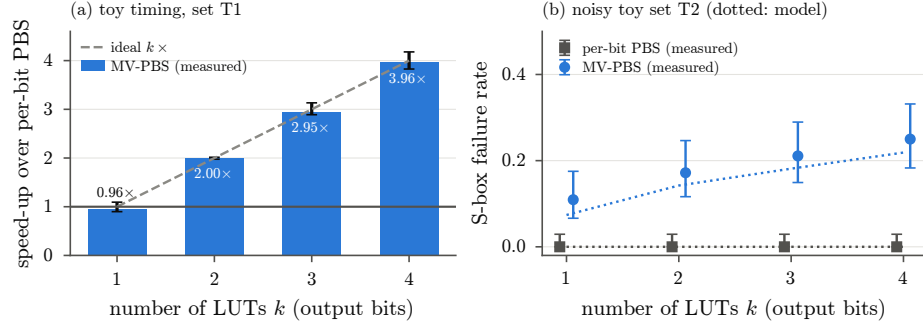


Fig. 1. (a) Speed-up of MV-PBS over per-bit PBS versus the number of tables k at T1, bars with paired-bootstrap confidence intervals; dashed: $k \times$. (b) S-box failure rate (any of the first k bits wrong) at T2 with Wilson intervals; dotted: Theorem 1.

i.e. 0.25 [0.183, 0.332]; Theorem 1 with an independence approximation predicts 0.219. So one blind rotation for all four bits is safe only when the parameters are chosen for the amplified noise.

A remark on CMux trees. Vertical packing [4] evaluates the whole table with four CMux gates, but only from GGSW encryptions of the input bits. From a single LWE ciphertext of the nibble this first requires extracting the bits and circuit-bootstrapping each of them, which in our toy costs more than one MV-PBS; we therefore do not report it as a competitor.

5 Conclusion

Sharing one blind rotation among the four output bits of a 4-bit S-box, as proposed in [2], gives a speed-up close to the number of bits in our toy, and the noise price is exactly the factor $\|d_f\|^2 \leq p + 2$ of Theorem 1. Two lessons of the falsification round carry over to real studies: speed-up claims must be checked at the edge of the claimed range (the scaling saturates), and comparisons must start from the same input format. *Limitations:* insecure toy parameters, pure-Python timings, and a single S-box; nothing here should be cited as a performance result.

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A Reproducing the Case Study

From the repository root:

```
cd case-study/toy-mvpbs
/usr/bin/python3 code/test_sbox_fhe.py
/usr/bin/python3 theory/check_mvpbs_noise.py
/usr/bin/python3 code/run_experiments.py
/usr/bin/python3 falsify/C3_linear_speedup/check_linear_speedup.py
/usr/bin/python3 code/make_figures.py
```

The ledger `EVIDENCE.md` maps every number of this paper to a log under `results/`; `CLAIMS.md` records which statements survived falsification.